

Step	Step Description	Agents Involved	Tool Calls and Parameters	Reason
1	User requests buildings in Exeter within 150m of a river	human, host_agent	Message from human to host_agent: "Show me buildings in Exeter that are within 150m of a river."	User initiates the query.
2	Plan steps to solve geospatial query	host_agent, planning_agent	send_message to planning_agent with task: "User wants to see buildings in Exeter that are within 150m of a river. Provide the general steps to solve this geospatial query."	host_agent delegates to planning_agent to break down the query into actionable steps.
3	Planning agent decomposes the query	planning_agent	Output: Steps are ["Find Exeter 1 area", "Find buildings in Exeter as many search results", "Find rivers in Exeter 1 or more results", "Apply condition of buildings within 150m of the river"]	Steps are based on the planning_agent_prompt, which instructs to identify area, entities, and apply spatial conditions.
4	Find area polygon for Exeter	host_agent, named_area_agent	send_message to named_area_agent: "Find the area polygon for Exeter."	named_area_agent is chosen because named_area contains polygons of areas; required to spatially bound the search.
5	Retrieve named area polygons	named_area_agent	call_os_ngd with polygon_or_point=True, filename='exeter_area_polygon'	named_area_agent uses the OS NGD tool to get all named area polygons, as per its prompt.
6	Filter for Exeter polygon	host_agent, data_analysis_agent	send_message to data_analysis_agent: "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save as 'filtered_exeter_area_polygon'."	Data analysis is needed because the NGD tool returns many polygons; filtering is required to isolate Exeter.
7	Generate metadata for artifact	data_analysis_agent	generate_metadata_for_artifacts for 'exeter_area_polygon'	Metadata is generated to understand the columns and values for filtering.
8	Code execution: filter Exeter polygon	coding_agent, code_executor	Code filters for rows where name1_text contains 'Exeter' and descriptiongroup/description indicate a settlement/city/town/suburban area. Output: 1 polygon found.	Filtering logic is based on metadata: name1_text and descriptiongroup/description columns are authoritative for identifying Exeter.
9	Confirm filtered Exeter polygon	data_analysis_agent, named_area_agent	Output: Artifact 'filtered_exeter_area_polygon' generated	Confirmation step; ensures correct polygon is isolated for further spatial queries.
10	Find all buildings in Exeter area	host_agent, buildings_agent	send_message to buildings_agent: "Find all buildings within the area defined by 'filtered_exeter_area_polygon'."	buildings_agent is chosen because it queries the buildings database; bbox parameter is set to the filtered Exeter polygon.
11	Retrieve buildings within Exeter	buildings_agent	call_os_ngd with bbox='filtered_exeter_area_polygon', filename='all_buildings_in_filtered_exeter_area'	The bbox parameter ensures only buildings within Exeter are returned, as per buildings_agent prompt.
12	Output building artifacts	buildings_agent	Output: Artifacts for buildings, building lines, and building parts generated	Multiple datasets are returned to cover all building representations.
13	Find all rivers in Exeter area	host_agent, water_network_agent	send_message to water_network_agent: "Find all rivers within the area defined by 'filtered_exeter_area_polygon'."	water_network_agent is chosen because it queries rivers and water networks; bbox is set to Exeter polygon.
14	Retrieve river features	water_network_agent	call_os_ngd with filters=['rivers'], bbox='filtered_exeter_area_polygon', filename='exeter_rivers'	Filters are set to 'rivers' to focus on relevant water features.
15	Output river artifacts	water_network_agent	Output: Artifacts for waterlinkset and waterlink generated	Both datasets are needed for comprehensive river representation.
16	Filter river features (waterlinkset)	host_agent, data_analysis_agent	send_message to data_analysis_agent: "Analyze 'waterlinkset_exeter_rivers' to filter out only the river data. Save as 'filtered_rivers_waterlinkset'."	Filtering is required because waterlinkset may contain non-river features; only river-related features are needed.
17	Generate metadata for river artifact	data_analysis_agent	generate_metadata_for_artifacts for 'waterlinkset_exeter_rivers'	Metadata is used to identify relevant columns for filtering (description, name1_text).
18	Code execution: filter river features (waterlinkset)	coding_agent, code_executor	Code filters rows where description or name1_text indicate river, brook, stream, leat, or watercourse. Output: 10 river features.	Filtering logic is based on metadata: description and name1_text columns are authoritative for identifying rivers.
19	Filter river features (waterlink)	host_agent, data_analysis_agent	send_message to data_analysis_agent: "Analyze 'waterlink_exeter_rivers' to filter out only the river data. Save as 'filtered_rivers_waterlink'."	Both waterlinkset and waterlink are filtered for completeness; waterlink may have more granular features.
20	Generate metadata for river artifact (waterlink)	data_analysis_agent	generate_metadata_for_artifacts for 'waterlink_exeter_rivers'	Metadata is used to identify relevant columns for filtering.
21	Code execution: filter river features (waterlink)	coding_agent, code_executor	Code filters rows where description or name1_text indicate river, brook, stream, leat, or watercourse. Output: 341 river features.	Filtering logic is based on metadata: description and name1_text columns are authoritative for identifying rivers.
22	Confirm filtered river artifacts	data_analysis_agent, water_network_agent	Output: Artifacts 'filtered_rivers_waterlinkset' and 'filtered_rivers_waterlink' generated	Confirmation step; ensures correct river features are isolated for spatial analysis.
23	Plot buildings within 150m of rivers	host_agent, plotting_agent	send_message to plotting_agent: "Plot all buildings from 'buildings_all_buildings_in_filtered_exeter_area' that are within 150m of any river from 'filtered_rivers_waterlink' in Exeter. Show spatial relationship and highlight buildings that meet the distance condition."	plotting_agent is chosen for spatial analysis and visualization; both buildings and filtered rivers artifacts are used.
24	Generate metadata for plotting	plotting_agent	generate_metadata_for_artifacts for 'buildings_all_buildings_in_filtered_exeter_area' and 'filtered_rivers_waterlink'	Metadata is used to ensure correct columns and CRS for spatial operations.
25	Code execution: plot buildings within 150m of rivers	coding_agent, code_executor	Code: Buffer rivers by 150m, find intersecting buildings, plot all layers on a folium map, summarize first 5 results. Output: 17,530 buildings within 150m, map artifact generated.	Buffering and intersection logic is based on spatial analysis best practices; columns used are geometry, osid, description, buildinguse, geometry_area_m2 as per metadata.
26	Output map and summary	plotting_agent, host_agent	Output: Map and summary of buildings within 150m of rivers in Exeter	Final output is a folium map and summary, as per plotting_agent_template.

Overall Summary: - Databases used: Named area (for Exeter polygon), buildings (for all buildings in Exeter), water network (for all rivers in Exeter). - Filters/code: Filtering for Exeter used name1_text and descriptiongroup/description columns; filtering for rivers used description and name1_text columns with keywords (river, brook, stream, leat, watercourse). Spatial analysis used a 150m buffer around river geometries and intersection with building polygons. - Final outcome: 17,530 buildings in Exeter are within 150 meters of a river. An interactive folium map was generated showing all buildings, rivers, the 150m buffer, and highlighted buildings within 150m of any river. - Overall reasoning: Each agent and tool was chosen based on its domain (named_area for area polygons, buildings for building data, water_network for rivers). Filtering steps were guided by authoritative metadata columns. Spatial analysis and visualization followed best practices for geospatial queries, ensuring accurate and relevant results for the user's request.